Bicol University
Bicol University Polangui
Polangui, Albay

Name: Sid Dione M. Molleda
Course & Section: BS Information System-2B
Subject: Information Management
Professor: Dr. Guillermo Red

Reflection on Using Document-Based NoSQL Database for User Activity Tracking in O'Clock E-Commerce Platform

In O'Clock, an e-commerce platform selling luxury watches, one component that could benefit from a document-based NoSQL database is tracking user activity, such as product browsing history and interactions. Traditional relational databases often face challenges when dealing with unstructured or semi-structured data, especially when user behavior and interactions are logged in varied formats. This is where NoSQL databases like MongoDB or Firebase shine, providing a flexible schema design that accommodates changes in data types over time without requiring extensive schema modifications.

For example, in O'Clock, users' browsing histories, product views, and clickstreams could be stored in a NoSQL database as JSON-like documents. Each document could represent a single user session, with fields that capture relevant details such as the products viewed, time spent on the platform, and the devices used. Unlike relational databases, which would require creating complex relational tables to join data from multiple sources, NoSQL systems can efficiently store all this information in one document, simplifying queries and enhancing performance.

Consider a local luxury watch store in the Philippines that wants to track customer interactions. By leveraging a NoSQL database, they could easily store dynamic data such as customer preferences, which may evolve over time, including the watches they viewed or added to their wish lists. This flexibility allows the store to quickly adjust to customer behavior trends and provide personalized experiences, such as product recommendations, without overhauling the database schema.

The use of a NoSQL database like MongoDB or Firebase would significantly improve the scalability of O'Clock, enabling the system to handle large amounts of data generated by user interactions while providing fast query capabilities. This is especially beneficial in an e-commerce environment, where responsiveness and adaptability are key to a positive user experience and business growth.

References:

Firebase. (n.d.). *Firebase Realtime Database*. https://firebase.google.com/docs/database
MongoDB. (n.d.). *What is MongoDB?*. https://www.mongodb.com/what-is-mongo